

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	2.78	7.42	0.56	1.12	1.17	0.32	6.21	4.42	3.14	7.71

Iteration 0: **step 0.**

Insertion Sort

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Iteration 1: **step 0.**

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Iteration 2: **step 0.**

Insertion Sort

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Value	0.56	2.78	7.42	1.12	1.17	0.32	6.21	4.42	3.14	7.71



Iteration 2: **step 1.**

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Value	0.56	2.78	7.42	1.12	1.17	0.32	6.21	4.42	3.14	7.71

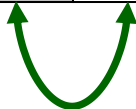
Iteration 2: **step 2.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	0.56	2.78	1.12	7.42	1.17	0.32	6.21	4.42	3.14	7.71



Iteration 3: **step 0.**

Insertion Sort

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Value	0.56	1.12	2.78	7.42	1.17	0.32	6.21	4.42	3.14	7.71



Iteration 3: **step 1.**

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Iteration 3: **step 2.**

Insertion Sort

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Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

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Value	0.56	1.12	2.78	1.17	7.42	0.32	6.21	4.42	3.14	7.71



Iteration 4: **step 0.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

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Value	0.56	1.12	1.17	2.78	7.42	0.32	6.21	4.42	3.14	7.71



Iteration 4: **step 1.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

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Value	0.56	1.12	1.17	2.78	7.42	0.32	6.21	4.42	3.14	7.71

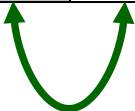
Iteration 4: **step 2.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

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Iteration 5: **step 0.**

Insertion Sort

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Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	0.56	1.12	1.17	0.32	2.78	7.42	6.21	4.42	3.14	7.71



Iteration 5: **step 1.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

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Iteration 5: **step 2.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
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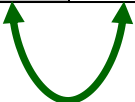
Iteration 5: **step 3.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

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Iteration 5: **step 4.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

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Iteration 5: **step 5.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

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Value	0.32	0.56	1.12	1.17	2.78	6.21	7.42	4.42	3.14	7.71



Iteration 6: **step 0.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	0.32	0.56	1.12	1.17	2.78	6.21	7.42	4.42	3.14	7.71

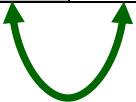
Iteration 6: **step 1.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	0.32	0.56	1.12	1.17	2.78	6.21	4.42	7.42	3.14	7.71



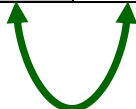
Iteration 7: **step 0.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	0.32	0.56	1.12	1.17	2.78	4.42	6.21	7.42	3.14	7.71



Iteration 7: **step 1.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	0.32	0.56	1.12	1.17	2.78	4.42	6.21	7.42	3.14	7.71

Iteration 7: **step 2.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	0.32	0.56	1.12	1.17	2.78	4.42	6.21	3.14	7.42	7.71



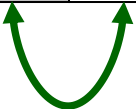
Iteration 8: **step 0.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	0.32	0.56	1.12	1.17	2.78	4.42	3.14	6.21	7.42	7.71



Iteration 8: **step 1.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	0.32	0.56	1.12	1.17	2.78	3.14	4.42	6.21	7.42	7.71



Iteration 8: **step 2.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	0.32	0.56	1.12	1.17	2.78	3.14	4.42	6.21	7.42	7.71

Iteration 8: **step 3.**

Insertion Sort

Iteration i . Repeatedly swap element i with the one to its left if smaller.

Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	0.32	0.56	1.12	1.17	2.78	3.14	4.42	6.21	7.42	7.71

Iteration 9: **step 0.**

Insertion Sort

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Property. After i th iteration, $a[0]$ through $a[i]$ contain first $i+1$ elements in ascending order.

Array index	0	1	2	3	4	5	6	7	8	9
Value	0.32	0.56	1.12	1.17	2.78	3.14	4.42	6.21	7.42	7.71

Iteration 10: **DONE.**